Newton's Method for Solving $f(x) = 0$

Dr. Allan Struthers

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The Root-Finding Problem

Goal: Find $x^* \in \mathbb{R}$ such that

$$f(x^*) = 0.$$

Assumptions:

- f is differentiable near x^*
- $f'(x^*) \neq 0$

Exact solutions are often unavailable, so we seek an **iterative numerical method**.

Linear Approximation of f

Let x_n be a current approximation to the root.

Using the first-order Taylor expansion at x_n:

$$f(x) \approx f(x_n) + f'(x_n)(x - x_n)$$

This is the equation of the **tangent line** to f at x_n.

Geometric Idea Behind Newton's Method

Key idea: Approximate the root of $f(x) = 0$ by the root of its tangent line.

Set the linear approximation equal to zero:

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$$

Solve for x_{n+1}.

Derivation of the Newton Update

From

$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = 0,$$

we obtain

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Newton's method iteration:

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

Algorithmic Form

Newton's Method:

1. Choose an initial guess x_0
2. For $n = 0, 1, 2, \ldots$ compute

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

3. Stop when $|x_{n+1} - x_n|$ or $|f(x_n)|$ is sufficiently small

Local Convergence Behavior

If $f \in C^2$ and x_0 is sufficiently close to a simple root x^*:

$$|x_{n+1} - x^*| \leq C|x_n - x^*|^2$$

Conclusion:

- Newton's method converges *quadratically*
- Accuracy roughly doubles at each iteration

When Newton's Method Can Fail

- Poor initial guess
- $f'(x_n) = 0$ or very small
- Multiple roots $(f'(x^*) = 0)$

In practice, Newton's method is often combined with:

- damping or line search
- bracketing methods (bisection)

Summary

- Newton's method uses a local linear approximation
- Each step solves the tangent line equation
- Fast (quadratic) convergence near simple roots
- Care needed for robustness

Newton's Method in the Complex Plane

Problem (Complex Version): Find $z^* \in \mathbb{C}$ such that

$$f(z^*) = 0,$$

where $f : \mathbb{C} \to \mathbb{C}$ is holomorphic.

Newton iteration:

$$\boxed{z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}}$$

Interpretation:

- $f'(z)$ is the complex derivative
- Each step solves the tangent (linear) approximation in $\mathbb{C}$
- Quadratic convergence near simple roots

New phenomena in $\mathbb{C}$:

- Basins of attraction with fractal boundaries
- Sensitivity to initial guesses
- Links to complex dynamics and Julia sets